

Wine Quality Exploratory Data Analysis

MBAV 443M – Statistical Modeling and AI-Driven Analytics

Assignment 1

Name: Divyam Mor | Roll No: 2528711

i. Introduction

Wine quality depends on a complex combination of physicochemical factors that alter sensory experiences. Finding out which chemical features correlate most with quality scores has a direct commercial impact on both producers and buyers. This paper conducts an exploratory data analysis (EDA) of a white wine dataset, which contains 4,898 samples, each characterized by eleven physicochemical variables - such as acidity, residual sugar, sulphates, and alcohol - Beyond a quality rating by an expert on a scale of 3 to 9. The goal is to create a profile of the data, evaluate its quality, discover distributional patterns, and reveal the characteristics most tied to wine quality, So laying the groundwork for subsequent predictive modelling.

ii. Exploratory Data Analysis

The dataset consists of 4,898 records with 12 variables in each. All cells are filled and there are no missing values. Out of the 12 columns, 11 are continuous numeric variables and one is a discrete integer score - quality. Duplicate detection resulted in 937 identical records which when removed, produced a clean working dataset of 3,961 unique observations. This significant duplication rate (about 19%) is quite a finding and could mean that the same wine samples were measured multiple times.

Distribution of quality scores is mostly centered on the middle: 1,175 wines got 5, 1,788 wines got 6 and 689 wines got 7. Combined they make up over 92% of the total data. Ratings on extremes are almost non-existent - only 20 wines scored 3 and 5 wines got the highest score of 9. Average quality is 5.88 and median 6.0, which is in line with the nearly normal but slightly right skewed quality distribution, so most wines can be considered 'average'. For the chemical components, alcohol averages 10.51% with a standard deviation of 1.23%, whereas residual sugar is the most spread out feature (mean 6.39 g/L, max 65.8 g/L), signifying the presence of outliers.

iii. Methodology

The analysis was conducted in Python using Pandas, NumPy, Matplotlib, and Seaborn. The workflow proceeded through five stages: (i) environment setup and library imports; (ii) data ingestion from a semicolon-delimited CSV file; (iii) structural inspection using `df.info()` and `df.describe()` to establish data types, shape, and summary statistics; (iv) data quality assessment covering missing values and duplicate detection; and (v) correlation analysis using Pearson coefficients computed across all numeric variables. Three visualisations were produced to support interpretation: a histogram of quality score frequencies, a boxplot comparing alcohol content across quality tiers, and a full correlation heatmap. All figures were exported at 150 DPI using `plt.savefig()`.

iv. Visualization

The results of the data analysis were made clear by creating three different charts. Figure 1 depicts how wine quality scores are spread out. It reconfirms that most of the wines have been rated as 5 and 6, with very few being rated extremely high or low. Figure 2 draws out a distinct positive correlation between the amount of alcohol and the quality of the wine: the median alcohol gradually increases from about 9.5% for quality-5 wines up to more than 12% for quality-8 wines. The correlation heatmap displayed in Figure 3 details all the pairwise relationships between features in a quantifiable manner.

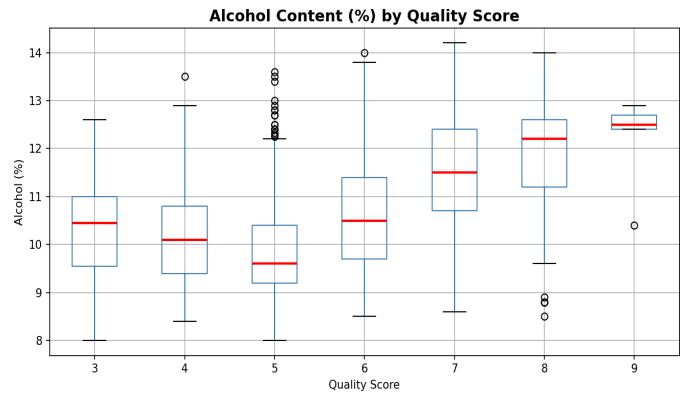


Figure 1 (left): Distribution of Wine Quality Scores — quality ratings 5 and 6 dominate the sample. Figure 2 (right): Alcohol Content (%) by Quality Score — higher-rated wines consistently show greater alcohol levels.

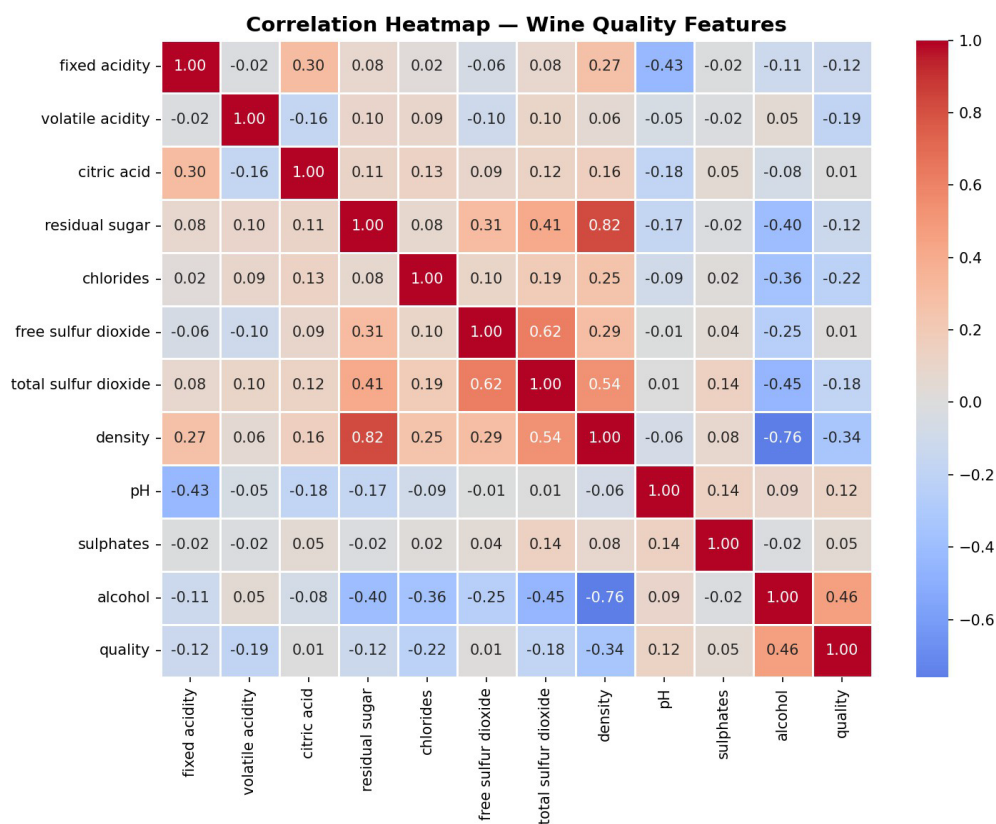


Figure 3: Correlation Heatmap — alcohol content shows the strongest positive correlation with quality (0.46), while density exhibits the most negative correlation (-0.34).

v. Key Insight

The most important and useful discovery is that there is a high positive correlation between the level of alcohol and the quality that people think the wine has ($r = 0.46$). This connection is very strong everywhere in the different levels of quality, which has been proven by Figure 2's boxplot: wines with a rating of 8 or 9 have a median alcohol content roughly 23 percentage points higher than those rated 5. However, the capacity of density to predict in the opposite direction is the most that can be countered with ($r = 0.34$), which is chemically understandable the denser wines are those that have more residual sugar and less alcohol, both of which are factors linked to lower scores.

The presence of volatile acidity and chlorides in a wine are also explainable by their significant negative correlations with wine quality ($r = 0.19$ and 0.22 respectively). This means that tasters penalize wines that have elevated off-flavor compounds and salt content.

In sum, the evidence based on these chemical indicators suggests that the main reason that determines the quality of white wine in this sample is the degree to which the fermentation has gone to completion and this is shown by the two wine parameters - high alcohol and low density.

vi. Conclusions

This EDA describes the analysis of a white wine dataset containing 3,961 unique samples characterized by 11 physicochemical features and a quality score. The data is clean, complete, and normally distributed with a median score of 6. Alcohol content is the strongest predictor of quality, with density (negatively), chlorides, and volatile acidity also having significant effects. Residual sugar and free sulphur dioxide correlate very weakly with quality, which means that they are unlikely to be good predictors on their own. Such results can be used as a guide to variable-selection for further regression or classification modelling where based on the results alcohol density, and volatile acidity would be expected to be the most powerful predictors. Further studies could investigate interactions among these variables and employ dimensionality reduction to find out whether an even smaller subset of features can still explain the variations in quality.

AI assistance (Claude) was used for editing and structuring this report. All analysis, interpretation, and conclusions are the author's own.

